CRYONANO

QuantumVolt Precision Isolated Voltage Source (4 / 8 / 16 Channel)

Low-Noise Filtering for Cryogenic Quantum & Measurement Systems

Precision Voltage Sourcing.
Isolated Low-Noise Outputs.
Built for Quantum Applications.

QuantumVolt is an ultra-low-noise, precision isolated voltage source designed for nanoelectronics, quantum transport, and quantum computing experiments.

It delivers stable and reproducible gate and bias control through fully isolated bipolar outputs. The architecture ensures minimal noise, low drift, and high channel isolation, enabling precise tuning of sensitive quantum devices across long-duration experiments.

Notes

- Outputs are current-limited and optimized for precision, not high-power delivery
- Designed specifically for low-noise experimental environments
- Custom channel counts and configurations available

KEY FEATURES

- **Isolated Multi-Channel Architecture**
  Available in 4, 8, and 16-channel configurations with fully isolated outputs scalable up to ±40 V for extended biasing.
- **Precision Voltage Control**
  Features ±10 V bipolar output per channel with 16-bit DAC resolution for fine voltage steps, smooth sweeps, and high-resolution tuning.
- **Ultra-Low Noise Performance**
  Optimized analog design minimizes output noise while preserving fragile quantum states and low-signal measurements.
- **Stability & Accuracy**
  Ensures high stability, low drift operation, and accurate bias conditions with minimal temperature dependence during long experiments.
- **Protection & Reliability**
  Integrated overvoltage (±30 V) and short-circuit protection ensure safe operation for sensitive devices and setups.

APPLICATIONS

- Quantum device gate control (qubits, superconducting circuits)
- Nanoelectronic and 2D material experiments
- Quantum transport measurements
- Multi-terminal device biasing
- Low-noise precision voltage sourcing
- Automated measurement systems

24-Channel Cryogenic Breakout Box

SPECIFICATIONS

OUTPUT SPECIFICATIONS

Output Voltage Range (Nominal)	±10 V (single fixed range), stackable to ±40 V
Number of Outputs	4 output lines (depending on device configuration)
Output Connectors	BNC sockets (standard)
Output Current per Output	Approx. current drive ±10 mA per channel typical
Output Resistance	100 mΩ
Programming Resolution	16 Bits
Crosstalk (Channel to Channel)	100 dB
Common Mode Voltage Rejection	120 dB
Slew Rate	7 V/µs
Noise (0.1 Hz – 10 kHz)	260 µVrms
Power-on Output State	High impedance

ACCURACY (MEASUREMENT CONDITIONS)

Parameter	Percent of Reading (Gain Error)	Percent of Range* (Offset Error)	Typical Drift Gain and Offset
Accuracy	0.01 %	0.1 %	6 ppm/°C , 80 µV/°C

Cryonano Labs | Head Office: Sector V, WTL Building, BP-5, BP Block, Sector V, Bidhannagar, Kolkata- 700091
Bangalore Office: #999/12, 2nd Floor, GMN Complex, 5th 'A' Cross, 1st Block, HRBR Layout, Babusapalya, Banaswadi, Outer Ring Road, Bangalore – 560043.
www.cryonano.com | Phone: 097481 81485

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